

LLM vs Human Discriminator — Real or Synthetic Tabular Data

A research framework to evaluate whether LLMs and humans can distinguish real from synthetic tabular data, across two information conditions.

Project Structure

```
llm_discriminator/  
├── data/  
│   ├── real/           ← place your real.csv here  
│   └── synthetic/       ← place your synthetic.csv here  
├── llm_discriminator/  
│   ├── metadata.py      ← derives distributional metadata from any dataframe  
│   ├── prompt_builder.py ← constructs C1 / C2 prompts  
│   └── evaluate.py       ← runs LLM discrimination trials (CLI)  
├── human_interface/  
│   └── app.py           ← Streamlit labeling UI for human participants  
├── analysis/  
│   └── analysis.ipynb    ← full results analysis notebook  
├── results/             ← auto-created; all .jsonl and .csv outputs land here  
├── envs/  
│   ├── environment.yml  ← conda env for SURF/DAS  
│   └── requirements.txt  
├── .env.example          ← copy to .env and add your API keys  
└── README.md
```

Setup

1. Environment (recommended: conda on SURF/DAS)

```
bash
```

```
# Clone / copy the project
cd llm_discriminator

# Create conda environment
conda env create -f envs/environment.yml
conda activate llm_discriminator

# OR use pip only
python -m venv .venv
source .venv/bin/activate
pip install -r envs/requirements.txt
```

2. API Keys

```
bash

cp .env.example .env
# Edit .env and fill in OPENAI_API_KEY and/or ANTHROPIC_API_KEY
```

3. Add your data

```
bash

cp /path/to/your/real_data.csv      data/real/real.csv
cp /path/to/your/synthetic_data.csv data/synthetic/synthetic.csv
```

Running the LLM Discriminator

```
bash
```

```
# Basic run – OpenAI GPT-4o, both conditions, 10 trials each
```

```
python -m llm_discriminator.evaluate \  
    --real data/real/real.csv \  
    --synthetic data/synthetic/synthetic.csv \  
    --provider openai \  
    --model gpt-4o \  
    --n_trials 10 \  
    --conditions C1 C2
```

```
# Anthropic Claude
```

```
python -m llm_discriminator.evaluate \  
    --real data/real/real.csv \  
    --synthetic data/synthetic/synthetic.csv \  
    --provider anthropic \  
    --model claude-opus-4-5-20251101 \  
    --n_trials 10
```

Results are written to `results/llm_results_<provider>_<timestamp>.jsonl`

Running the Human Interface

```
bash
```

```
streamlit run human_interface/app.py
```

- Opens at <http://localhost:8501>
- Participants enter their ID, choose conditions, and judge tables one by one
- Results are appended to `results/human_results.jsonl` in real time

Deploying on SURF/DAS (port forwarding)

```
bash
```

```
# On the compute node:
```

```
streamlit run human_interface/app.py --server.port 8501 --server.address 0.0.0.0
```

```
# On your local machine (SSH tunnel):
```

```
ssh -L 8501:localhost:8501 <username>@<surf-node-address>
```

```
# Then open http://localhost:8501 in your browser
```

Running the Analysis Notebook

```
bash

cd analysis
jupyter notebook analysis.ipynb
# OR on SURF/DAS:
jupyter nbconvert --to notebook --execute analysis.ipynb --output analysis_executed.
```

Experimental Conditions

Condition	What the evaluator sees
C1	Raw table (first N rows)
C2	Raw table + full distributional metadata (mean, std, skewness, kurtosis, top values, correlations, Shapiro-Wilk normality, etc.)

Output Files

File	Description
results/llm_results_<provider>_<ts>.jsonl	LLM trial-level results
results/human_results.jsonl	Human trial-level results (all participants)
results/final_summary.csv	Aggregated accuracy + significance tests
results/*.png	Analysis plots

Extending the Study

- **Add a synthesizer:** Generate a new synthetic CSV with SDV (`sdv.SingleTableMetadata` + any synthesizer) and drop it in `data/synthetic/`
- **Add a model:** Add a new entry to `PROVIDER_MAP` in `evaluate.py`
- **Add a condition C3:** Extend `prompt_builder.py` with a new template and add `"C3"` to the CLI options

- **Multi-dataset:** Loop `run_experiment()` over multiple `(real_path, synthetic_path)` pairs
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Citation (when publishing)

If you use this framework, please cite the relevant synthesizer papers (CTGAN, GReaT, REaLTabFormer) and the LLM-as-evaluator literature (Zheng et al. 2023).